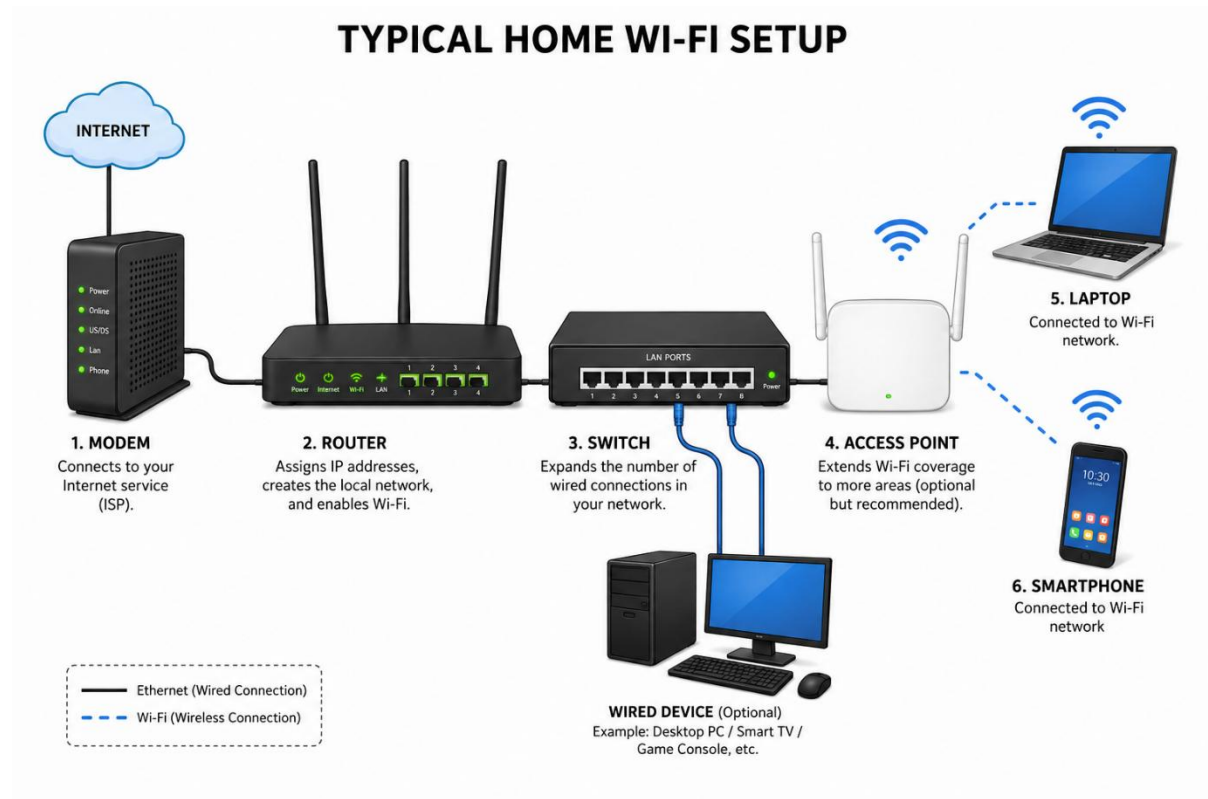


# Module 3 (Assignment Day-1)

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').



| Device            | Main Function                                                                   | Real-Life example                                                                         |
|-------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Modem             | Connects your home network to the Internet Services Provider (ISP).             | Used at home to receive internet services from JIO, Airtel, etc.                          |
| Router            | Share the internet connection between many devices and manages network traffic. | Used at home to connect phone, laptops, and smart TV to WI-FI.                            |
| Switch            | Connects multiple wired devices together on the same network.                   | Used in an office to connect several computer using ethernet cables.                      |
| Access Point (AP) | Provide or extend WI-FI coverage in an area.                                    | Used in a large house, colleges, or office to provide WI-FI in rooms far from the router. |
| Network Cable     | Carries data between networking devices and computers using a wired connection. | Ethernet cable used to connect a PC to a router or switch.                                |

**3.**Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



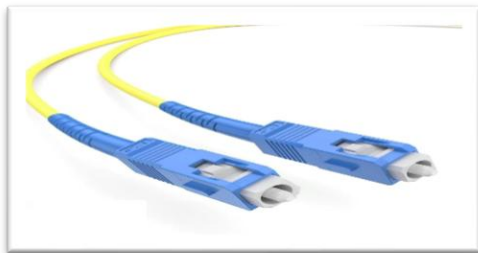
**1) Ethernet Cables:**



**2) Coaxial Cables:**



**3) Fiber Optic Cables:**



| <b>Cables Type</b>        | <b>Common Use</b>                             |
|---------------------------|-----------------------------------------------|
| <b>Ethernet Cables</b>    | Home and Office LAN connections.              |
| <b>Coaxial Cables</b>     | Cable TV and Cable internet.                  |
| <b>Fiber Optic Cables</b> | High-speed and Long-distance internet cables. |

4.Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.



| <b>Difference</b>          | <b>Router</b>                                                         | <b>Switch</b>                                                      |
|----------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>Main use</b>            | Connects different networks, such as a home network to the Internet.  | Connects multiple devices within the same network.                 |
| <b>Typical location</b>    | Usually placed between the modem and the devices in a home or office. | Usually placed inside an office, computer lab, or data center.     |
| <b>How it handles data</b> | Sends data between different networks using <b>IP addresses</b> .     | Sends data to the correct device using <b>MAC addresses</b> .      |
| <b>Devices connected</b>   | Connects a local network to the Internet or another network.          | Connects computers, printers, servers, and other devices on a LAN. |
| <b>Internet connection</b> | Usually provides or manages Internet access.                          | Normally does not directly provide Internet access.                |
| <b>Example</b>             | Home Wi-Fi router connecting your phone and laptop to the Internet.   | Network switch connecting 20 computers in a computer lab.          |